MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC ASSESSMENT – 10** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 8: Introduction to Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 Multiple Choice Questions and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal subparts.

SECTION A: Objective Type Questions (6 × 1 = 6 Marks)

Q1. For any acute angle θ, the value of $\sin^2\theta + \cos^2\theta$ is always: [1]

(A) Less than 1 (B) Equal to 1
(C) Greater than 1 (D) Depends on θ

Q2. As the acute angle θ increases from 0° to 90°, the value of $\cos\theta$: [1]

(A) Increases from 0 to 1 (B) Decreases from 1 to 0
(C) Remains constant (D) Increases then decreases

Q3. If $\tan\theta + \cot\theta = 2$, then the value of $\tan^{10}\theta + \cot^{10}\theta$ is: [1]

(A) 1 (B) 2 (C) 2^{10} (D) 20

Q4. If $x = 2\sin^2\theta$ and $y = 2\cos^2\theta + 1$, then the value of $x + y$ is: [1]

(A) 2 (B) 3 (C) 1 (D) 5

Q5. If $\triangle ABC$ is an equilateral triangle, then the value of $\sin A + \cos B$ is: [1]

(A) $\frac{\sqrt{3}+1}{2}$ (B) $\sqrt{3}$
(C) 1 (D) $\frac{\sqrt{3}}{2}$

Q6. **Assertion (A):** If $\sec\theta + \tan\theta = 5$, then $\sec\theta - \tan\theta = 0.2$.
Reason (R): $\sec^2\theta - \tan^2\theta = 1 \implies (\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = 1$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \operatorname{cosec} 30^\circ}$. [2]

Q8. If $\tan\theta + \cot\theta = 2$, find the value of $\tan^2\theta + \cot^2\theta$. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Prove that: $\frac{\cot A - \cos A}{\cot A + \cos A} = \frac{\csc A - 1}{\csc A + 1}$. [3]

Q10. If $\tan\theta = \frac{20}{21}$, evaluate $\frac{1 - \sin\theta + \cos\theta}{1 + \sin\theta + \cos\theta}$. [3]

SECTION D: Long Answer / HOTS (2 × 5 = 10 Marks)

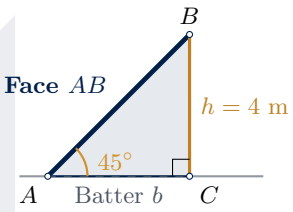
Q11. If $\csc\theta - \sin\theta = l$ and $\sec\theta - \cos\theta = m$, prove that $l^2 m^2 (l^2 + m^2 + 3) = 1$. [5]

Q12. If $x = r\sin A\cos C$, $y = r\sin A\sin C$, and $z = r\cos A$, prove that $r^2 = x^2 + y^2 + z^2$. [5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Retaining Wall Earth Embankment Batter Slope

A civil engineering department constructs a retaining wall with a front batter face inclined to prevent earth slippage. The cross-section forms a right-angled triangle where the vertical wall height is $h = 4$ m, and the face is sloped at an angle of $\theta = 45^\circ$ to the horizontal foundation bed.



Based on the above geometric model, answer the following questions:

(i) State the value of $\tan 45^\circ$ and find the horizontal batter spread b of the foundation. [1]

(ii) Determine the length of the diagonal inclined face of the retaining wall. [1]

(iii) Verify that $\sin^2 45^\circ + \cos^2 45^\circ = 1$ using the dimensions of this structure. [2]

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★★★ END OF QUESTION PAPER · ALL THE BEST ★★★